

Assignment 3: Understanding Token Log Probabilities - Solution

Objective: Create a Python program that accepts a prompt, generates a GPT response, retrieves token log probabilities, and displays token statistics.

Python Code

```
from openai import OpenAI
import math

client = OpenAI(api_key="YOUR_API_KEY")

prompt = input("Enter your prompt: ")

response = client.responses.create(
    model="gpt-4.1-mini",
    input=prompt,
    logprobs=True
)

print("\nGenerated Response:\n")

cumulative_probability = 1.0

for item in response.output:
    if hasattr(item, "content"):
        for content in item.content:
            if hasattr(content, "text"):
                print(content.text)

            if hasattr(content, "logprobs") and content.logprobs:
                print("\nToken Analysis:")
                print("-"*60)
                print(f'{"Token":15}{"LogProb":15}{"Probability":15}{"Cumulative"}')

                for token in content.logprobs:
                    log_prob = token.logprob
                    probability = math.exp(log_prob)
                    cumulative_probability *= probability

                    if probability > 0.80:
                        color = "GREEN"
                    elif probability > 0.50:
                        color = "YELLOW"
                    else:
                        color = "RED"

                print(f'{"token.token":15}{"log_prob:<15.4f"}{"probability:<15.4f"}{"cumulative_probability:.6f"} {color}')
```

Explanation

1. Accepts a prompt from the user.
2. Generates a response using a GPT model.
3. Retrieves the log probability of each generated token.
4. Converts log probability to probability using $\exp(\log_prob)$.
5. Computes cumulative probability.
6. Color codes tokens:
 - GREEN: Probability > 0.80
 - YELLOW: Probability > 0.50
 - RED: Probability ≤ 0.50